

Fine-tuning the Wav2Vec2 Model for Automatic Speech Emotion Recognition System

Devendra Kayande, Indra Ballav Sonowal and Ramesh K. Bhukya, *IEEE Senior Member*

Department of Electronics and Communication Engineering

Indian Institute of Information Technology Allahabad, Prayagraj, Uttar Pradesh - 211015, India

{iec2021088, iec2021114 and rkbhukya}@iita.ac.in

Abstract—The speech emotion recognition (SER) has gained pivotal attention on various applications in human-computer interaction and affective computing. In these days, there has been a growing interest in developing robust and accurate systems for identifying emotions from speech utterances. In this work, a novel approach to Wav2Vec2 architecture is used to demonstrate the SER system performance. The Wav2Vec2 model is used to extract the speech features from utterances and feed to feed forward network to identify the emotions accurately on the two datasets, namely, *Toronto emotional speech set (TESS)* and *Crowd-sourced Emotional Multimodal Actors Dataset (CREMA-D)*. Wav2Vec2 implements a contrastive learning target during the pre-training stage. The CREMA-D achieved an accuracy of 76%. Additionally, the weighted F1 score, precision, and recall were, respectively, 0.76, 0.77, and 0.77. On the other hand, on the TESS dataset achieved an accuracy of 99%, the F1 score was 0.99. Furthermore, the weighted recall and precision were both 0.99 and 0.99.

Index Terms—Speech Emotion Recognition, Wav2Vec2, Human-Computer Interaction

I. INTRODUCTION

Speech emotion recognition (SER) refers to the ability to identify and understand human emotions through various cues such as facial expressions, vocal tone, body language, and physiological responses [1]. This field has gained significant attention in recent years due to its potential applications in different areas of everyday life, medical use, and various industries [2]. SER can enhance communication by providing individuals with insights into the emotional state of others, enabling them to respond appropriately. It can be used in customer service to gauge customer satisfaction and tailor responses accordingly. SER can assist in mental health diagnosis and treatment by providing objective data on patients' emotional states [3]. It can be used to monitor individuals with conditions like *autism spectrum disorder*, helping caregivers understand their emotional needs and provide appropriate support. SER is gaining popularity in industries such as healthcare, retail, and entertainment [4].

A survey conducted by the AI-based SER conference found that 81% of respondents believed that emotion recognition technology would become mainstream within the next five years. In recent studies, several approaches have been explored to address emotion ambiguity. Fayek et al. [5], introduced a soft-target label, while Lotfian et al. [6] treated the presence of multiple emotions in a single utterance as a multi-task

problem. Ando et al. [7] and Fei et al. [8] employed a multi-label approach to capture the nuances of emotion ambiguity. An investigation conducted by Beard et al. [9] focused on the Crema-D dataset and utilized COVAREP features along with an LSTM decoder. This study achieved an audio-only accuracy of 41.5% across eight emotions. Similarly, Jin et al. [10] used the same dataset, but with six emotions, and developed ranking-based SVM classifiers for SER, resulting in an overall accuracy of 51.82%. Gokilavani et. al., [11] used custom *convolutional neural networks (CNNs)* on RAVDESS, CREMA and TESS dataset. Basu et al. [12] published a brief review on the importance of speech emotion datasets and features, noise reduction; ultimately, they analyze the significance of different classification approaches, including SVM and HMM. Ahmad et al., [13] used ensemble models of LSTMs and CNNs with additional data augmentation by adding noise and achieved accuracy around 90% on small SER datasets like TESS and SAVEE. Valles et al. [14] conducted a sentiment analysis, employing ensemble learning for emotion recognition. Among their various models, the RNN model yielded an accuracy of 55%. Additionally, Singh et al. [] conducted binary classification analysis on each individual emotion within the Crema-D dataset. In their research, Shukla et al. [15] utilized generative text based techniques with GRUs and achieved an accuracy of 55.01% on the CREMA-D dataset. Cai et al. [16] introduced a purely attention-based network called audio spectrogram transformer, resulting in an accuracy of 67.81%. Yingzhi et al. [17] proposed a partial fine-tuning approach using HuBERT for emotion classification, which yielded an accuracy of 78.91%. Itai et al. [18] proposed a gradient-based adversarial learning framework that focused on normalizing speaker characteristics within the feature representation for SER. Their method achieved an accuracy of 81% on the IEMOCAP dataset. Astuti et al. [19] introduced a method for feature extraction utilizing Gaussian MFCCs.

In [20] incorporates the RAVDESS dataset involving the extraction of three essential features, namely Mel Frequency Cepstral Coefficients (MFCCs), Mel Spectrogram, and chromagram presents the development of the *Voice Driven Emotion Recognizer Mobile Phone (VDERM)* system for automatic SER from speech [21]. It outlines the design, feature extraction using LPC analysis, selection of 18 speech features for emotion representation, and the creation of a diverse database for training and recognition. The study compares neural network

and fuzzy model recognition methods, achieving a 60% recognition rate. In [22] employed a *Deep Neural Network* (DNN) approach for SER, utilizing convolutional, pooling, and fully connected layers. The study focuses on a 3-class subset (angry, neutral, sad) from the German-based Emo-DB corpus which were standardized and segmented into 20-millisecond non-overlapping segments using *voice activity detection* (VAD) [23]–[25]. The optimized DNN model, trained without feature selection, achieved an impressive 96.97% overall test accuracy. Sahu et al [26] introduces SER system that differs from traditional models by predicting emotions for multiple segments within a single audio clip. The system utilizes a pre-trained deep CNN followed by a single-layered neural network to classify emotion classes for these audio segments. It achieves a notable accuracy of 68.7%, surpassing current state-of-the-art models when applied to the IEMOCAP audio-only dataset for classifying emotions into four categories *angry*, *happy*, *sad*, and *neutral*. In [27] employed a custom CNN plus MFCCs extraction method with a custom dataset with RAVDESS outperforming some transfer learning tasks. In [28] explores the application of semi-supervised ladder networks for speech emotion recognition. The model is trained through the minimization of both supervised loss and auxiliary unsupervised cost functions with experiments conducted on the Interactive Emotional Dyadic Motion Capture (IEMOCAP) database. Han et al., [29] introduces parallel network composed of ResNet, CNN, and transformer encoder components. Experiments on the RAVDESS dataset confirm the method's effectiveness. The audio files were subjected to feature extraction using Wav2Vec2 [30], [31], yielding context representations. These representations underwent transformations through successive attention heads, culminating in the derivation of a 768-dimensional vector. Subsequently, this vector was processed by a dense neural network classifier comprised of 128 units, featuring dropout, and terminated with a final softmax layer. The overall pipeline was applied to address this multiclass classification problem posed by the raw audio speech files. Wav2Vec2 excels in extracting nuanced features from raw audio signals, making it a powerful tool for speech recognition [32]. Its self-supervised learning approach enables pre-training on extensive unlabeled datasets, enhancing its ability to capture intricate details of spoken language. In real-world applications, Wav2Vec2 is utilized for tasks such as transcription services, voice assistants, and audio indexing [33]–[35]. Wav2Vec2's adaptability to multilingual scenarios underscores its significance in advancing automatic speech recognition technologies.

In this paper, we delve into the intricate details of our research, which is organized into six sections. Section II provides a comprehensive description of the dataset used in our study, Section III, highlighting the data preprocessing and feature extraction methods. In Section IV, elucidate the intricacies of our model architecture, showcasing how Wav2Vec2 is tailored to serve as the backbone of our SER system. Section V is dedicated to the presentation of results and an in-depth discussion of our findings, offering insights into the system's performance and its implications. Finally, in Section VI,

conclude our paper, summarizing our contributions, discussing potential future work, and emphasizing the significance of advancing SER technology.

II. DATABASE

CREMA-D The CREMA-D (Crowd-sourced Emotional Multimodal Actors Dataset) is a widely used dataset in the field of speech emotion recognition. It consists of a collection of audio and video recordings of professional actors performing scripted sentences with different emotional expressions. The dataset contains a total of 7,442 audio recordings. The dataset involves the participation of 91 professional actors (48 males and 43 females) from various age groups and diverse ethnic backgrounds. The dataset covers six distinct emotional categories: *angry*, *disgust*, *fear*, *happy*, *neutral*, and *sad*. The dataset was divided in 80%-20% manner into train and test data. The histogram of the CREMA-D dataset can be observed from the Fig. 1

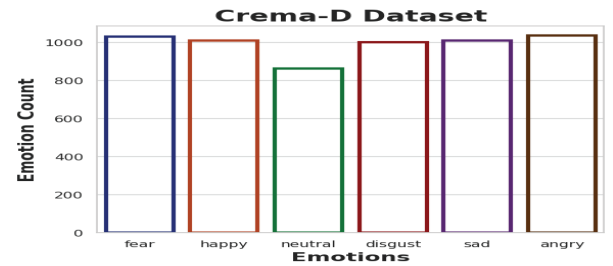


Fig. 1. The number of speech files present in each emotions are shown.

TESS The TESS datasets consist of audio recordings where professional actors were instructed to vocalize specific emotions, including anger, disgust, fear, happiness, pleasant surprise, sadness, and neutral expressions. Each audio file corresponds to a specific emotion category and is associated with an actor ID. The audio files are typically in a standardized audio format (WAV) and have consistent audio quality. TESS dataset contains 2800 audio files, with 400 examples for each of the seven target emotions. The recordings are 3 to 5 seconds long and were produced using a professional studio setup. The TESS dataset from Fig. 2 shown the number of speech files present in each emotion. Both of the datasets are fairly balanced.

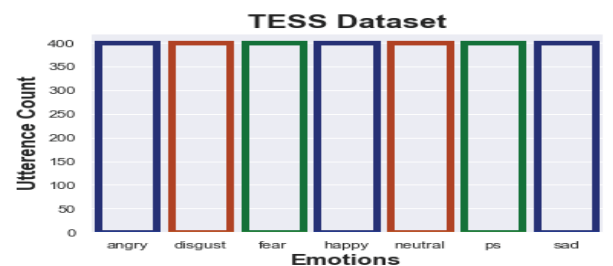


Fig. 2. The number of speech files present in each emotions are shown.

III. METHODOLOGY

In order to preprocess the audio into classification model, there is a need to set up the relevant Wav2Vec2 assets. The Wav2Vec2 model extracts 3D features from the audio waveform which are nothing but the output of transformer layers, including:

- Time-domain Representations(dimension = 0,time axis): The model initially operates in the time domain and processes the raw audio waveform directly. It employs a series of convolutional layers to capture local temporal patterns and dependencies in the audio signal.
- Mel-filterbank Energies(dimension =1, frequency/energy axis): The Wav2Vec2 model utilizes a mel-filterbank to convert the time-domain signal into a frequency-domain representation. It divides the audio signal into multiple frequency bands and computes the energy or magnitude in each band using mel filters. This process yields a sequence of mel-filterbank energies, which provide information about the distribution of energy across different frequency bands over time.
- Perceptually Relevant Features(dimension =2, channel axis): Wav2Vec2 further enhances the representation by incorporating features that are perceptually relevant for speech. It employs a convolutional neural network (CNN) to capture contextual information within the audio input. To handle the context representations in any audio length a merge strategy used to concatenate 3D representations generated by Wav2Vec2 feature extractor into 2D feature vector representations.

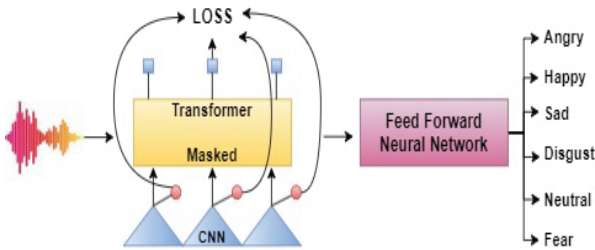


Fig. 3. Feature extraction process for extracting contextual information from speech utterances.

There are three merge strategies: mean, sum, and max. The better results can be achieved on the mean(averaging along dimension = 1 , the frequency dimension) approach. By taking the mean along dimension = 1, calculating the average frequency content at each moment in time. This can be useful for analyzing the overall spectral characteristics or energy distribution of the audio signal over time. It provides a way to summarize the frequency information and create a simplified representation of the audio's spectral content.

The Wav2Vec2 model requires audio files with a sampling rate of 16000 Hz hence initially resampled all the audio files to 16kHz sampling rate. The features extracted from the audio path in context representation 2D tensors and feed them into classification model to determine the emotion in the speech.

IV. MODEL TRAINING AND TESTING

Model: Wav2Vec2 is a speech recognition model that was proposed by Facebook for learning speech representations in the context of *automatic speech recognition* (ASR). The model is trained on a massive dataset of unlabeled speech audio, and it learns to represent the speech signal in a way that is useful for downstream tasks such as speech recognition and speaker identification. Wav2Vec2 is based on the CNN and Transformer architecture , which is a neural network architecture characterized by its use of self-attention, which allows the model to learn long-range dependencies in the input data. The architecture of Wav2Vec 2.0 is composed of three main parts:

Feature encoder: This is the first part of the model, and its job is to reduce the dimensionality of the audio data. It does this by converting the raw waveform into a sequence of feature vectors, each of which represents a 25 millisecond window of audio. The feature encoder is a 7-layer convolutional neural network with 512 channels at each layer which processes the audio signal in a hierarchical manner to capture both local and global acoustic features.

Transformer: The transformer is the second part of the model, and its job is to learn long-range dependencies in the audio data. It does this by attending to different parts of the feature vector sequence, and learning how to combine them to represent the overall audio signal. The transformer is a stack of 12 encoder and decoder layers, with 8 attention heads per layer.

Output layer: The output layer is the final part of the model, and its job is to generate the final prediction. It does this by taking the output of the transformer and passing it through a linear layer, which produces a sequence of probabilities that represent the likelihood of each possible output.

Wav2Vec2 is trained using a self-supervised learning task called masked language modeling. In masked language modeling, the model is given a sequence of tokens, and some of the tokens are masked out. The model is then trained to predict the masked tokens. This task forces the model to learn to represent the speech signal in a way that captures the statistical relationships between the different tokens in the sequence. During this pre-training phase, Wav2Vec2 uses a contrastive learning objective. This involves masking portions of the input audio and training the model to predict the masked portions using the remaining context. By learning to reconstruct the masked segments, the model is encouraged to capture meaningful representations that capture relevant acoustic information. Wav2Vec2 employs vector quantization to further compress the feature representations. The continuous feature vectors are quantized into a discrete set of codewords, reducing the memory footprint of the model while maintaining important information. The training objective is a sum of two loss functions: contrastive loss and diversity loss

$$L_T = L_c + \alpha L_D \quad (1)$$

The contrastive loss (L_C) plays a crucial role in training the model to accurately predict the quantized representation,

denoted as q_t from a set Q_t . This set Q_t , comprises the target representation q_t and $K + 1$ other distractor representations q' sampled uniformly from different time steps in the mask. A constant temperature parameter, k is used throughout the training process. To calculate the loss, cosine similarities between the context representations c_t and the quantized representation q are computed.

$$L_m = -\log \frac{e^{\text{sim}(\frac{c_t, q_t}{k})}}{\sum_{\bar{q} \in Q_t} e^{\text{sim}(\frac{c_t, \bar{q}}{k})}} \quad (2)$$

Diversity Loss: is a regularization technique that utilizes entropy and can be computed using the following formula.

$$H(X) = -\sum_x P(x) \log(P(x)) \quad (3)$$

where x - a possible outcome of a discrete random variable X , $P(x)$ - probability of event x .

Encouraging the model to make use of all code words is achieved by maximizing the entropy, which is essentially equivalent to minimizing the negative entropy, known as diversity loss.

$$L_d = \frac{1}{GV} * (-H(\bar{p}_g)) = \frac{1}{GV} \sum_{g=1}^G \sum_{v=1}^V \bar{p}_{g,v} \log(\bar{p}_{g,v}) \quad (4)$$

In our work, the pretrained Wav2Vec2 model is utilized from the Hugging Face model hub. Specifically from the speech classification checkpoint. During the fine-tuning process, the weights of the feature extractor component were frozen. This means that the feature extractor did not receive gradient updates while training.

To build the speech classification model, a small feed-forward network is constructed, consisted of a dense layer with 128 units, followed by a dropout layer with a dropout rate of 0.2. The 768-dimensional vector generated by the Wav2Vec2 feature extractor was passed as input to feed-forward network. Finally, a softmax layer was added at the end to produce the classification probabilities for our target classes. For CREMA-D 6 units softmax was applied and for TESS dataset 7 units softmax was applied as the datasets are categorised into 6 and 7 classes respectively.

Training and Testing Phase During the training phase, a data collator is employed to dynamically pad the tokens generated by Wav2Vec2 of the audio data. The data collator handles the input values and labels differently, applying separate padding functions to each of them. This differentiation is essential as speech input and output belong to distinct modalities and should be treated with distinct padding functions. To ensure this, the labels are padded with -100 , making them disregarded during loss computation, similar to how other data collators handle padding tokens. This utilization of separate padding functions guarantees appropriate treatment of input and output in the context of different modalities.

For the CREMA-D and TESS dataset nearly the same training procedure was used. The training process involved the

utilization of specific hyperparameters. The learning rate was set to $3e^{-5}$, as a very high learning rate easily paves the way for overfitting for such large models, while the batch size for training and evaluation was set to 4. To accumulate gradients, a gradient accumulation step of 2 was employed, resulting in an effective total batch size of 8. The optimization algorithm used was *Adam*, with β values set at 0.9 and 0.999, and an ϵ value of $1e^{-8}$. A linear learning rate scheduler was employed, and the training was conducted for a total of 4 epochs for CREMA-D as longer training on small datasets of these large models can easily lead to overfitting and degradation of model performance.

V. RESULTS AND DISCUSSION

During the evaluation phase, the performance of the model was assessed on a total of 1489 number of samples from the CREMA-D dataset and 560 samples from the TESS dataset. The inclusion of these datasets are used to evaluate the model's generalization and effectiveness across different datasets and gather a comprehensive understanding of its performance. The test data was processed in batches to ensure efficient handling of the audio files. Preprocessing techniques, such as tokenizing, were then applied to adequately prepare the data for the fine-tuned model. Following tokenization, the preprocessed audio files were passed through the fine-tuned model for classification. The fine-tuned model has undergone specialized training to adapt its parameters to the specific task at hand, which in this case is the classification of emotions in audio data. By leveraging the knowledge learned during the training phase, the model is able to make predictions on the test data.

	angry	disgust	fear	happy	neutral	sad
angry	214	11	1	7	2	0
disgust	20	196	17	8	4	24
fear	13	9	156	12	7	42
happy	18	8	21	198	14	1
neutral	2	2	4	10	201	5
sad	0	25	42	5	21	169
	angry	disgust	fear	happy	neutral	sad

Fig. 4. With the exception of sadness, which is often considered as neutral, the group's perception of an emotion tends to match the intended emotions closer than other emotions, according to the confusion matrix between intended emotion and group perception.

In this multiclass classification task, the system evaluation is employed using precision, recall, F1 score, and accuracy considered as performance metrics. These metrics provided a comprehensive assessment of the model's performance. Analysis of the confusion matrix in Figure 4 reveals that the model demonstrates confusion primarily among the emotions of *sad*, *fear*, and *disgust*. Specifically, certain audio files labeled as *sad* are misclassified as either *disgust* or *fear*. Moreover, the model also misclassified some *angry* audio

TABLE I

The classification report on the TESS and CREMA-D dataset. The precision, recall, F1-score, accuracy, macro average and weighted average are the performance metrics taken into account for the SER system evaluation.

Dataset	TESS				CREMA-D			
Parameters/ Emotions	Precision	Recall	F-1 score	Support	Precision	Recall	F-1 score	Support
Angry	1.00	1.00	1.00	101	0.80	0.91	0.85	235
Disgust	0.99	0.98	0.98	108	0.78	0.73	0.75	269
Fear	1.00	0.99	0.99	94	0.65	0.65	0.65	239
Happy	0.99	0.99	0.99	107	0.82	0.76	0.79	260
Neutral	1.00	1.00	1.00	102	0.81	0.90	0.85	224
Pleasant Surprise	0.99	0.98	0.98	89				
Sad	0.97	1.00	1.00	99	0.70	0.65	0.67	262
Accuracy	99%				76%			
Macro Average	0.99	0.99	0.99	700	0.76	0.77	0.76	1489
Weighted Average	0.99	0.99	0.99	700	0.76	0.76	0.76	1489

files as *sad*. This behavior can be attributed to the shared characteristics or features present among these emotions. The observed misclassifications imply that the distinctions between the emotional categories of *sad*, *fear*, and *disgust* may not be well drawn, making it challenging to make the distinctions. Similarly, the misclassification of *angry* audio files as *sad* suggests the presence of overlapping acoustic characteristics between these emotions.

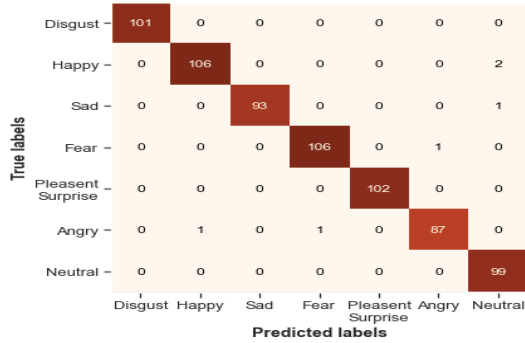


Fig. 5. The group perception much more closely matches the intended emotions, compared to the confusion matrix between intended emotion and group perceived emotion.

According to Fig. 5, the fine tuned model performed well on the TESS dataset, resulting in minimal of 6 misclassification. Overall, the presence of maximum data in the strong diagonal of the classification matrix shows that the model performs well in classifying the labels.

Table I displays the accuracy, precision, recall, and F1 score for both the CREMA-D and TESS datasets. Additionally, we have also displayed the macro average and weighted average. According to Table I, 76% accuracy was attained on the CREMA-D dataset. The macro average precision, recall, and F1 score are respectively 76%, 77%, and 76%. Additionally, the precision, recall, and F1 score weighted averages are 76%, 76%, and 76%, respectively. On the other hand, the accuracy achieved in the TESS dataset is 99%. Precision, recall, and F1 scores all fall within the macro average range of 99%, 99%,

and 99%. Additionally, the weighted averages for F1 score, precision, and recall are 99%, 99%, and 99%, respectively.

VI. CONCLUSION AND FUTURE DIRECTION

In this research work we investigated the latest technique for SER. By developing robust and effective methods using these large transformer architecture models for SER, aimed to contribute to advancements in these areas. The Wav2Vec2 model is a transformer-based pre-trained on vast amounts of unlabeled audio data, which allows it to learn general representations and extract meaningful features from speech signals. This pre-training helps capture a wide range of acoustic characteristics and enables the model to effectively encode speech information, including emotional cues. After pre-training, the Wav2Vec2 model can be fine-tuned on a labeled dataset specific to SER. This transfer learning approach can lead to improved accuracy and efficiency in recognizing emotions from speech data. This motivated us to use this large audio model for further analysis.

Based on the evaluation, the results obtained on the CREMA-D dataset achieved is 76%, and on the TESS dataset 99%. Additionally, the macro and weighted average of precision, recall, and F1 score computed. The precision, recall, and F1 score achieved on the CREMA-D dataset are respectively 76%, 77%, and 76% while the precision, recall, and F1 score weighted averages are 76%, 76%, and 76%, respectively. On the TESS dataset, the macro average precision, recall, and F1 score are 99%, 99%, and 99% respectively and the weighted average precision, recall, and F1 score are also 99%, 99%, and 99%, respectively.

In near future, the research in SER may focus on large-scale annotated datasets. Transformer architectures play a pivotal role in enabling effective fusion, context modeling, attention visualization, and continual adaptation. Based on the recent trends, the SER systems may be deployed in real-time scenarios and foster their applicability in various domains, including human-computer interaction, affective computing, and psychological research.

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